

Day 2 – Querying & Modifying Data (29/12/25)

1.

```
create database insuranceDB;
```

```
Commands completed successfully.
```

2.

```
--customers table
```

```
create table customers (  
    customerId int primary key,  
    firstName varchar(10) null,  
    lastName varchar(50) not null,  
    dateOfBirth date not null,  
    phone varchar(12) not null,  
    email varchar(50) not null  
);
```

```
--policies table
```

```
create table policies(  
    policyId int primary key,  
    policyName varchar(20),  
    policyType varchar(20),  
    premiumAmount decimal(10,2),  
    durationYears int  
)
```

```
--claims table
```

```
create table claims(  
    claimId int primary key,  
    assignmentId int,  
    claimDate date,  
    claimAmount decimal(10,2),  
    claimStatus varchar(5),  
    foreign key(assignmentId) references policyAssignments(assignmentId)  
)
```

```
--agents table
```

```
create table agents(  
    agentId int primary key,  
    agentName varchar(50),  
    phone varchar(12),  
    city varchar(50)  
)
```

```
--policy assignments
```

```
create table policyAssignments(  
    assignmentId int primary key,  
    policyId int,  
    agentId int,  
    claimId int,  
    claimDate date,  
    claimAmount decimal(10,2),  
    claimStatus varchar(5),  
    foreign key(policyId) references policies(policyId),  
    foreign key(agentId) references agents(agentId),  
    foreign key(claimId) references claims(claimId)
```

```

assignmentId int primary key,
customerId int,
policyId int,
agentId int,
startDate date,
endDate date
foreign key (customerId) references customers(customerId),
foreign key (policyId) references policies(policyId),
foreign key (agentId) references agents(agentId)
)

```

3.

--insert customers

```

insert into customers values(1,'Amit','Sharma','1995-10-10','9989727567','amit@mail.com'),
(2,'preethi','varma','1998-09-08','8989898989','preethi@mail.com')

```

--insert agents

```

insert into agents values(101,'Santhosh','9989898988','Hyd')

```

--insert policies

```

insert into policies values(120,'Health Secure','Health',12000.00,1),
(121,'Life Shield','Life',25000.00,10),
(122,'Motor Protect','Motor',8000.00,1)

```

--insert policy assignments

```

insert into policyAssignments values(202,1,120,101,'2020-10-10','2020-12-10')

```

--insert claims

```

insert into claims values(1000,202,'2020-10-10',10000,'Yes')

```

4.

1. select * from customers;

	customerId	firstName	lastName	dateOfBirth	phone	email
1	1	Amit	Sharma	1995-10-10	9989727567	amit@mail.com
2	2	preethi	varma	1998-09-08	8989898989	preethi@mail.com
3	3	Vishwa	Nair	2014-10-10	8976598765	vishwa@mail.com

2. select customerId,policyId,startDate,endDate from policyAssignments;

	customerId	policyId	startDate	endDate
1	1	120	2020-10-10	2020-12-10
2	2	120	2021-09-10	2021-10-12
3	1	121	2021-09-10	2021-12-11

3. select * from policies where policyType='Health';

	policyId	policyName	policyType	premiumAmount	durationYears
1	120	Health Secure	Health	12000.00	1

4. select * from policies where premiumAmount>10000 and durationYears=1;

	policyId	policyName	policyType	premiumAmount	durationYears
1	120	Health Secure	Health	12000.00	1

5. select distinct city from agents;

	city
1	Delhi
2	Hyderabad
3	Kakinada

6. select * from policies where policyType='Health' or policyType='Life' or policyType='Motor';

	policyId	policyName	policyType	premiumAmount	durationYears
1	120	Health Secure	Health	12000.00	1
2	121	Life Shield	Life	25000.00	10
3	122	Motor Protect	Motor	8000.00	1

7. select * from policies where policyType in ('Health','Life','Motor');

	policyId	policyName	policyType	premiumAmount	durationYears
1	120	Health Secure	Health	12000.00	1
2	121	Life Shield	Life	25000.00	10
3	122	Motor Protect	Motor	8000.00	1

8. select * from customers where dateOfBirth>='2001-1-1' and dateOfBirth<='2020-12-31';

	customerId	firstName	lastName	dateOfBirth	phone	email
1	3	Vishwa	Nair	2014-10-10	8976598765	vishwa@mail.com

9. select * from customers where dateOfBirth between '2001-1-1' and '2020-12-31';

	customerId	firstName	lastName	dateOfBirth	phone	email
1	3	Vishwa	Nair	2014-10-10	8976598765	vishwa@mail.com

10. select * from claims where claimStatus='Rejected';

	claimId	assignmentId	claimDate	claimAmount	claimStatus
1	1000	202	2020-10-11	10000.00	Rejected

11. select * from agents where city like '_a%';

	agentId	agentName	phone	city
1	104	Hithesh	8338334564	Kakinada

12. select max(claimAmount) as maxClaimAmount,min(claimAmount) as minClaimAmount from claims;

	maxClaimAmount	minClaimAmount
1	20000.00	10000.00

13. select max(claimDate) as latestDate from claims;

	latestDate
1	2020-12-11

14. update policies set premiumAmount=premiumAmount*1.1 where policyType='Health';
select * from policies where policyType='Health';

	policyId	policyName	policyType	premiumAmount	durationYears
1	120	Health Secure	Health	13200.00	1

15. select * from policyAssignments;

	assignmentId	customerId	policyId	agentId	startDate	endDate
1	201	2	120	101	2020-10-10	2021-11-11
2	202	1	120	101	2020-10-10	2026-12-10
3	205	5	122	103	2025-12-12	2026-12-12
4	206	5	121	102	2024-11-10	2025-12-31
5	207	5	120	102	2025-11-12	2026-11-17

delete from policyAssignments where endDate<=GETDATE();
select * from policyAssignments;

	assignmentId	customerId	policyId	agentId	startDate	endDate
1	202	1	120	101	2020-10-10	2026-12-10
2	205	5	122	103	2025-12-12	2026-12-12
3	206	5	121	102	2024-11-10	2025-12-31
4	207	5	120	102	2025-11-12	2026-11-17

(1 row affected)

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16. select count(*) as noOfClaimsRejected from claims where claimStatus='Rejected';

	noOfClaimsRejected
1	1

17. select policyId, policyName, premiumAmount, premiumAmount * 0.06 as localTaxes,
premiumAmount + (premiumAmount * 0.06) as premiumAmountWithTax, (premiumAmount +
(premiumAmount * 0.06)) / 12 as monthlyPremiumAmount from policies;

	policyId	policyName	premiumAmount	localTaxes	premiumAmountWithTax	monthlyPremiumAmount
1	120	Health Secure	14520.00	871.2000	15391.2000	1282.6000000
2	121	Life Shield	25000.00	1500.0000	26500.0000	2208.3333333
3	122	Motor Protect	8000.00	480.0000	8480.0000	706.6666666

18.alter table customers add address varchar(50);
alter table customers add city varchar(20);

Commands completed successfully.

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	customerId	firstName	lastName	dateOfBirth	phone	email	address	city
1	1	Amit	Sharma	1995-10-10	9989727567	amit@mail.com	NULL	NULL
2	2	preethi	varma	1998-09-08	8989898989	preethi@mail.com	NULL	NULL
3	3	Vishwa	Nair	2014-10-10	8976598765	vishwa@mail.com	NULL	NULL

19.

alter table agents add DevOfId int;
select * from agents;

	agentId	agentName	phone	city	DevOfId
1	101	Santhosh	9989898988	Hyderabad	NULL
2	102	Ravi	9765898756	Delhi	NULL
3	103	Girish	8978789789	Hyderabad	NULL
4	104	Hithesh	8338334564	Kakinada	NULL

20. alter table agents add constraint rfk_devofid_agentid foreign key (DevOfId) references agents(agentId);

Commands completed successfully.

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	agentId	agentName	phone	city	DevOfId
1	101	Santhosh	9989898988	Hyderabad	NULL
2	102	Ravi	9765898756	Delhi	NULL
3	103	Girish	8978789789	Hyderabad	NULL
4	104	Hithesh	8338334564	Kakinada	NULL

5.

1. select policyName as policiesForCustomer5 from policies join policyAssignments on policies.policyId=policyAssignments.policyId where customerId=5;

	policiesForCustomer5
1	Motor Protect
2	Life Shield

- select concat(firstName,lastName) as name, policyName from customers join policyAssignments on customers.customerId=policyAssignments.customerId join policies on policyAssignments.policyId=policies.policyId;

	name	policyName
1	AmitSharma	Health Secure
2	JulienGou	Motor Protect
3	JulienGou	Life Shield

- select concat(firstName,lastName) as name, claimId,claimAmount,claimStatus from claims join policyAssignments on claims.assignmentId=policyAssignments.assignmentId join customers on policyAssignments.customerId=customers.customerId;

	name	claimId	claimAmount	claimStatus
1	AmitSharma	1000	10000.00	Rejected
2	JulienGou	1001	20000.00	Accepted

- select firstName, policyName, agentName, startDate, endDate from customers join policyAssignments on customers.customerId=policyAssignments.customerId join agents on policyAssignments.agentId=agents.agentId join policies on policies.policyId=policyAssignments.policyId;

	firstName	policyName	agentName	startDate	endDate
1	Amit	Health Secure	Santhosh	2020-10-10	2026-12-10
2	Julien	Motor Protect	Girish	2025-12-12	2026-12-12
3	Julien	Life Shield	Ravi	2024-11-10	2025-12-31

- select firstName, policyName, claimAmount, claimStatus, claimDate from claims join policyAssignments pa on pa.assignmentId=claims.assignmentId join customers on pa.customerId=customers.customerId join policies on policies.policyId=pa.policyId;

	firstName	policyName	claimAmount	claimStatus	claimDate
1	Amit	Health Secure	10000.00	Rejected	2020-10-11
2	Julien	Motor Protect	20000.00	Accepted	2023-11-12

- select firstName, policyName from customers left join policyAssignments pa on pa.customerId=customers.customerId left join policies p on p.policyId=pa.policyId;

	firstName	policyName
1	Amit	Health Secure
2	preethi	NULL
3	Vishwa	NULL
4	Hansy	NULL
5	Julien	Motor Protect
6	Julien	Life Shield

7. select firstName from customers left join policyAssignments as pa on pa.customerId=customers.customerId left join claims c on c.assignmentId=pa.assignmentId where pa.assignmentId is null;

	firstName
1	preethi
2	Vishwa
3	Hansy

8. select c.customerId,concat(c.firstName,c.lastName) as fullName,sum(claimAmount) from customers c join policyAssignments pa on c.customerId=pa.customerId join claims cl on cl.assignmentId=pa.assignmentId group by c.customerId,c.firstName,c.lastName;

	customerId	fullName	totalClaimAmount
1	1	AmitSharma	10000.00
2	5	JulienGou	75000.00

9. select c.customerId,concat(c.firstName,c.lastName) as fullName,sum(claimAmount) as totalClaimAmount from customers c join policyAssignments pa on c.customerId=pa.customerId join claims cl on cl.assignmentId=pa.assignmentId group by c.customerId,c.firstName,c.lastName having sum(claimAmount)>50000;

	customerId	fullName	totalClaimAmount
1	5	JulienGou	75000.00

10. select pa.agentId,agentName,count(*) as policyCount from agents join policyAssignments pa on pa.agentId=agents.agentId group by pa.agentId,agentName;

	agentId	agentName	policyCount
1	101	Santhosh	1
2	102	Ravi	2
3	103	Girish	1

